

ADVANCED DIGITAL COMMUNICATION

Lab Report: Bandwidth Requirement and Pulse Shaping

Roll No.: 22EC5003

Bit Rate Used: 3 Mbps (last two digits = 03)

1. Aim

To calculate the bandwidth required to transmit a 3 Mbps data stream using 16-QAM and QPSK, for power-containment factors $\gamma = 0.99$ and $\gamma = 0.90$, with (i) ideal sinc pulses and (ii) raised-cosine pulses with roll-off factor $\beta = 0.5$; and to plot the time-domain and frequency-domain representations of rectangular, sinc and sine pulses.

2. Software Used

- Python 3 (Jupyter Notebook)
- NumPy: numerical arrays and FFT
- SciPy: numerical integration (quad) and root finding (brentq)
- Matplotlib: plotting

3. Theory

3.1 Symbol rate

For an M-ary modulation scheme, each symbol carries $\log_2(M)$ bits, so the symbol rate is

$$R_s = R_b / \log_2(M)$$

16-QAM carries 4 bits/symbol and QPSK carries 2 bits/symbol. With $R_b = 3$ Mbps:

- 16-QAM: $R_s = 3 / 4 = 0.75$ Msymbols/s
- QPSK: $R_s = 3 / 2 = 1.5$ Msymbols/s

3.2 Power-containment bandwidth (γ)

The bandwidth required is defined as the bandwidth that contains a fraction γ of the total signal power:

$$\int \text{from } -B/2 \text{ to } B/2 \text{ of } |P(f)|^2 df = \gamma \cdot \int \text{from } -\infty \text{ to } \infty \text{ of } |P(f)|^2 df$$

where $P(f)$ is the spectrum of the pulse and B is the (RF) bandwidth.

3.3 Sinc pulse

The sinc pulse $p(t) = \text{sinc}(t/T_s)$ has an ideal rectangular spectrum of width $1/T_s = R_s$ (from $-R_s/2$ to $+R_s/2$). Since the power is uniformly distributed over this band, the bandwidth containing a fraction γ of the power is simply

$$B = \gamma \cdot R_s$$

3.4 Raised-cosine pulse

With the frequency variable normalised as $v = f \cdot T_s$, the raised-cosine spectrum with roll-off β is

$$P(v) = 1 \text{ for } |v| \leq (1-\beta)/2$$

$$P(v) = 0.5 [1 + \cos(\pi (|v| - (1-\beta)/2) / \beta)] \text{ for } (1-\beta)/2 < |v| < (1+\beta)/2$$

$$P(v) = 0 \text{ otherwise}$$

The total power is the integral of $|P(v)|^2$, and the normalised one-sided cut-off q is found by solving

$$\int \text{from } 0 \text{ to } q \text{ of } |P(v)|^2 dv = \gamma \cdot \int \text{from } 0 \text{ to } (1+\beta)/2 \text{ of } |P(v)|^2 dv$$

The RF bandwidth is then $B = 2 \cdot q \cdot R_s$. The root q is found numerically using Brent's method.

3.5 Rectangular, sinc and sine pulses

- **Rectangular pulse:** $p(t) = 1$ for $|t| \leq T/2$. Its spectrum is $T \cdot \text{sinc}(fT)$, with first null at $f = 1/T$ and slowly decaying side lobes.
- **Sinc pulse:** $p(t) = \text{sinc}(t)$. Its spectrum is an ideal rectangle of width 1 Hz ($|f| \leq 0.5$ Hz), but the pulse is non-causal and of infinite duration.
- **Sine pulse:** $p(t) = \sin(2\pi f_0 t)$. Its spectrum consists of impulses at $\pm f_0$ (here $f_0 = 5$ Hz).

4. Source Code

Q1: Sinc pulse bandwidth

```
import numpy as np

# Roll number = 22EC5003
# Last two digits = 03
Rb = 3.0 # Mbps

# Bits per symbol
bits_16qam = 4
bits_qpsk = 2

# Symbol rates
Rs_16qam = Rb / bits_16qam
Rs_qpsk = Rb / bits_qpsk

# Gamma values
gammas = [0.99, 0.90]

print("Q1: Sinc Pulse")
print("Bit Rate =", Rb, "Mbps")
print()

for gamma in gammas:

    # Sinc pulse: RF bandwidth = gamma x symbol rate
    BW_16QAM = gamma * Rs_16qam
    BW_QPSK = gamma * Rs_qpsk

    print("Gamma =", gamma)
    print("16-QAM Bandwidth =", BW_16QAM, "MHz")
    print("QPSK Bandwidth =", BW_QPSK, "MHz")
    print()
```

Q2: Raised-cosine pulse bandwidth ($\beta = 0.5$)

```
import numpy as np
from scipy.integrate import quad
from scipy.optimize import brentq
```

```

# Bit rate
Rb = 3.0 # Mbps

beta = 0.5

# Symbol rates
Rs_16qam = Rb / 4
Rs_qpsk = Rb / 2

# Raised cosine frequency response
def RC(v):

    v = abs(v)

    a = (1 - beta) / 2
    b = (1 + beta) / 2

    if v <= a:
        return 1.0

    elif v < b:
        return 0.5 * (1 + np.cos(np.pi * (v - a) / beta))

    else:
        return 0.0

# Total signal power
total_power = quad(
    lambda v: RC(v)**2,
    0,
    (1 + beta) / 2
)[0]

# Find bandwidth for a particular gamma
def find_bandwidth(gamma):

    def power(q):
        return quad(
            lambda v: RC(v)**2,
            0,
            q
        )[0]

    q = brentq(
        lambda q: power(q) / total_power - gamma,
        0,
        (1 + beta) / 2
    )

    return q

# Gamma values
for gamma in [0.99, 0.90]:

    q = find_bandwidth(gamma)

    BW_16QAM = 2 * q * Rs_16qam

```

```

BW_QPSK = 2 * q * Rs_qpsk

print("Gamma =", gamma)
print("16-QAM Bandwidth =", BW_16QAM, "MHz")
print("QPSK Bandwidth   =", BW_QPSK, "MHz")
print()

```

Q3: Time and frequency domain plots

```

import numpy as np
import matplotlib.pyplot as plt

# Sampling
Fs = 1000
T = 1

t = np.arange(-3, 3, 1/Fs)

# 1. Rectangular Pulse
rect = np.where(np.abs(t) <= T/2, 1, 0)

# 2. Sinc Pulse
sinc = np.sinc(t)

# 3. Sine Pulse
f_sine = 5
sine = np.sin(2 * np.pi * f_sine * t)

# Fourier Transform
N = len(t)
f = np.fft.fftfreq(N, 1/Fs)
RECT_F = np.fft.fftshift(np.fft.fft(rect)) / Fs
SINC_F = np.fft.fftshift(np.fft.fft(sinc)) / Fs
SINE_F = np.fft.fftshift(np.fft.fft(sine)) / Fs

# Plot
plt.figure(figsize=(12, 8))

# Time Domain
plt.subplot(2, 1, 1)
plt.plot(t, rect, label="Rectangular")
plt.plot(t, sinc, label="Sinc")
plt.plot(t, sine, label="5 Hz Sine")
plt.title("Q3: Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.xlim(-2, 2)
plt.grid()
plt.legend()

# Frequency Domain
plt.subplot(2, 1, 2)
plt.plot(f, np.abs(RECT_F), label="Rectangular")
plt.plot(f, np.abs(SINC_F), label="Sinc")
plt.plot(f, np.abs(SINE_F), label="5 Hz Sine")
plt.title("Q3: Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("|P(f)|")
plt.xlim(-10, 10)
plt.grid()
plt.legend()

plt.tight_layout()
plt.show()

```

5. Procedure

1. Take the last two digits of the roll number 22EC5003, i.e. 03, so the bit rate is $R_b = 3$ Mbps.
2. Compute the symbol rates $R_s = R_b/4$ for 16-QAM and $R_s = R_b/2$ for QPSK.
3. Sinc pulse: compute $B = \gamma \cdot R_s$ for $\gamma = 0.99$ and 0.90 for both modulation schemes.
4. Raised-cosine pulse: define the RC spectrum for $\beta = 0.5$, integrate $|P(v)|^2$ to get the total power, and use Brent's root-finding method to obtain q for each γ . Compute $B = 2 \cdot q \cdot R_s$.
5. Generate rectangular, sinc and 5 Hz sine signals over $t = -3$ s to 3 s with a sampling frequency of 1000 Hz.
6. Compute their Fourier transforms using the FFT (with fftshift and scaling by $1/F_s$) and plot the time-domain and frequency-domain responses.
7. Record the results in the tables and compare.

6. Observations

Given: $R_b = 3$ Mbps, $R_s(16\text{-QAM}) = 0.75$ Msps, $R_s(\text{QPSK}) = 1.5$ Msps.

Table 1: Bandwidth required with sinc pulse (MHz)

	16-QAM	QPSK
$\gamma = 0.99$	0.7425	1.4850
$\gamma = 0.90$	0.6750	1.3500

Table 1: Sinc pulse, $B = \gamma \cdot R_s$

Table 2: Bandwidth required with raised-cosine pulse, $\beta = 0.5$ (MHz)

	q (normalised)	16-QAM	QPSK
$\gamma = 0.99$	0.5570	0.8355	1.6711
$\gamma = 0.90$	0.4234	0.6352	1.2703

Table 2: Raised-cosine pulse, $B = 2 \cdot q \cdot R_s$ (total power of $|P(v)|^2$ for one side = 0.4375)

Pulse plots

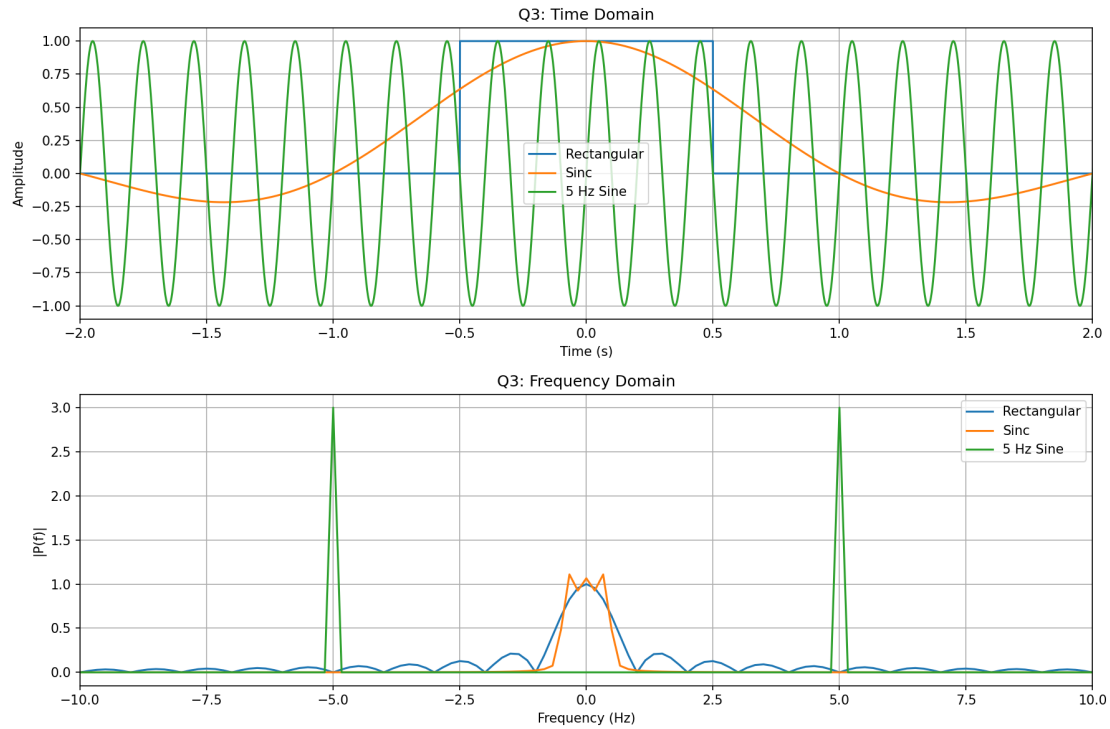


Figure 1: Time-domain and frequency-domain representation of rectangular, sinc and 5 Hz sine pulses

7. Results & Discussion

- **Effect of modulation order:** 16-QAM needs half the bandwidth of QPSK for the same bit rate (e.g. 0.7425 MHz vs 1.485 MHz for $\gamma = 0.99$ with sinc), because it carries twice as many bits per symbol and therefore has half the symbol rate.
- **Effect of γ :** a lower power-containment factor needs less bandwidth. Reducing γ from 0.99 to 0.90 reduces the sinc bandwidth by about 9%, and the raised-cosine bandwidth by about 24%, since the RC spectrum tapers and its tail carries very little power.
- **Sinc vs raised cosine:** for the same γ the raised-cosine pulse requires more bandwidth at $\gamma = 0.99$ (0.8355 MHz vs 0.7425 MHz for 16-QAM) because its spectrum extends to $(1+\beta)R_s/2$. At $\gamma = 0.90$ the RC pulse needs less bandwidth than the sinc pulse (0.6352 vs 0.6750 MHz), since the RC power is concentrated near the centre of the band. In return, the RC pulse has far faster-decaying time tails, which reduces inter-symbol interference sensitivity to timing errors and makes it practically realisable.
- **Rectangular pulse:** time-limited, so its spectrum is a sinc extending to infinity with side lobes. First null is at 1 Hz.
- **Sinc pulse:** band-limited with an almost flat spectrum within $|f| \leq 0.5$ Hz (the small ripple in the plot is due to the finite observation window of 6 s), but its time response extends infinitely.
- **Sine pulse:** the spectrum shows two sharp peaks at ± 5 Hz (magnitude 3 = amplitude $\times$ window length / 2), confirming the signal is a single-frequency tone.

Note: rectangular and sinc pulses are duals of each other; time-limited pulses are not band-limited and band-limited pulses are not time-limited.

8. Conclusion

The bandwidth for a 3 Mbps stream was computed for 16-QAM and QPSK using sinc and raised-cosine ($\beta = 0.5$) pulses at $\gamma = 0.99$ and 0.90 . Higher-order modulation (16-QAM) halves the required bandwidth compared to QPSK, and relaxing γ reduces it further. The raised-cosine pulse trades a little extra bandwidth at high γ for practical realisability and low ISI. The time and frequency plots verified the Fourier-pair relation between rectangular and sinc pulses and the line spectrum of the sine pulse.